

Table 5. Dataset categories and their descriptions

Dataset Name	Dataset Description
Comparative Relational Facts	Direct comparison, quantitative comparison, temporal comparison, multi-attribute comparison, ordinal ranking
Causal Chain Statements	X causes Y, causal chains with mechanisms, multi-step causation, feedback loops, preventive causation
Conditional Truths	If-then conditionals, when-then temporals, unless exceptions, necessary/sufficient conditions, biconditionals
Temporal Sequence Facts	Sequential ordering, comparative temporal, simultaneous events, duration statements, cyclic patterns
Quantitative Precision	Direct numerical assertions, comparative quantitative, multi-unit conversions, proportional relationships
Definitional & Taxonomic	X is defined as Y, X is a type of Y, taxonomic hierarchies, necessary conditions, process definitions
Negation & Absence Claims	Simple negation, modal impossibility, universal negatives, absence assertions, exception handling
Multi-Hop Inference Chains	Transitive relations, categorical syllogisms, hypothetical chains, if-then chains, compositional inference
Probabilistic & Hedged	Modal auxiliaries, probability adverbs, hedging phrases, quantified generalizations, statistical correlations
Process & Procedural	Linear sequential, cyclical processes, branching conditionals, phase transitions, multi-stage transformations
Control	A control setup where we have no fact at all and simply extract activations from the model control tokens